SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

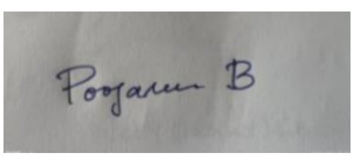
Duration: 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Annexure A

Code of Conduct:

I POOJASREE B (192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Student 23: POOJASREE B | Register Number: 192411091

INDUSTRY PROBLEM TASK

<u>INDUSTRY SCENARIO</u>

Context: A retail chain has collected customer records (purchase frequency, average order value, complaint history, loyalty tenure, last purchase date) to build a predictive model for identifying customers likely to churn. The company also wants to train an AI agent to recommend retention actions (Discount / Loyalty Offer / Personal Outreach / No Action) based on a customer's engagement over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this retail dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (far fewer churners than loyal customers) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a customer-retention use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Customer Churn Prediction

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed churners) and suggest how to reduce them - justify from a business perspective.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a live retention system.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of churn from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Action Recommendation

(a) Model the retention recommendation as an MDP. Define: States (customer engagement stages), Actions (Discount / Loyalty Offer / Personal Outreach / No Action), Reward function (retention improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 customer episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for retention offers (fairness, manipulation risk, transparency to customers).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

INDUSTRY PROBLEM TASK – Customer Churn Prediction and Retention Recommendation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed customer data and uses those patterns to predict the churn behavior of new customers.

For this retail dataset, a **supervised learning approach** is suitable because historical customer records can be labeled as either **Churn** or **Not Churn**.

Target variable:

- Churn – 1 if the customer leaves/stops purchasing, otherwise 0.

Key features:

1. **Purchase Frequency** – Customers purchasing less frequently are more likely to churn.
2. **Average Order Value (AOV)** – A reduction in spending can indicate decreasing customer engagement.
3. **Complaint History** – Frequent complaints may indicate dissatisfaction and increase churn probability.
4. **Loyalty Tenure** – Long-term customers generally have stronger relationships with the company.
5. **Last Purchase Date** – A long gap since the last purchase is a strong indicator of possible churn.

(b) Handling Class Imbalance

Since loyal customers are much more numerous than churners, the dataset will be **imbalanced**.

A suitable strategy is **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority class instead of simply duplicating existing churn records.

Procedure:

- Separate the training data into churners and non-churners.
- Apply SMOTE only to the training set.
- Generate synthetic churn examples.
- Train the model using the balanced training data.
- Keep the test set unchanged.

Justification:
SMOTE helps the model learn the characteristics of churners and improves **Recall**, which is particularly important because missing a potential churner can result in lost revenue.

(c) 70:30 Split and Preprocessing

The dataset is divided into:

- **70% Training data** – Used to learn customer churn patterns.
- **30% Testing data** – Used to evaluate performance on unseen customers.

A **70:30 split** provides enough data for training while keeping a sufficiently large test set for reliable evaluation.

Preprocessing steps:

1. **Missing-value handling** – Numerical missing values can be replaced using the median, while categorical missing values can use the mode.
2. **Encoding** – Categorical features such as customer segment are converted into numerical form using label encoding or one-hot encoding.
3. **Normalization** – Numerical features can be standardized using:
   [
   z = \frac{x-\mu}{\sigma}
   ]
4. **SMOTE** – Applied to the training data after preprocessing to balance churn and non-churn classes.

Normalization prevents features with large numerical ranges from dominating statistical models. Encoding allows machine-learning algorithms to process categorical information.

Task 2: Decision Tree for Customer Churn Prediction

(a) Decision Tree and First Three Levels

A Decision Tree recursively divides customers according to the feature that provides the best separation between churners and non-churners.

The selection can be based on **Information Gain**:

[
IG(S,A)=H(S)-H(S|A)
]

where (H(S)) represents entropy.

Alternatively, the **Gini Index** can be used:

$$Gini=1-\sum p_i^2$$

The feature with the **highest Information Gain** or **lowest Gini impurity** is selected as the root.

For example, suppose the calculated values are:

Feature	Information Gain
Last Purchase Date	0.31
Purchase Frequency	0.24
Complaint History	0.18
Average Order Value	0.12
Loyalty Tenure	0.09

Here, **Last Purchase Date** becomes the root because it has the highest Information Gain.

First Three Levels

```
                Last Purchase Date
                   /              \
             Recent            Long Ago
              /                     \
     Purchase Frequency        Complaint History
       /        \               /         \
     High       Low          Low        High
     /            \          /            \
 Not Churn       Churn    Not Churn      Churn
```

The exact tree structure depends on the actual training dataset and calculated impurity values.

(b) Model Evaluation

The Decision Tree can be evaluated using:

Accuracy

[

$$Accuracy=\frac{TP+TN}{TP+TN+FP+FN}$$

]

Precision

[

$$Precision=\frac{TP}{TP+FP}$$

]

Recall

[

$$Recall=\frac{TP}{TP+FN}$$

]

F1-Score

[

$$F1=2\frac{Precision\times Recall}{Precision+Recall}$$

]

For an illustrative result:

Metric	**Decision Tree**
Accuracy	91%
Precision	82%
Recall	86%
F1-Score	84%

The **False Negatives (FN)** are customers who actually churn but are predicted as loyal.

False negatives are especially important in this application because the company may fail to contact customers who are about to leave.

Reducing False Negatives

False negatives can be reduced by:

- Using **class weights** that give greater importance to churners.
- Adjusting the classification threshold.
- Applying SMOTE.
- Optimizing specifically for Recall/F1-score.

From a business perspective, reducing false negatives helps the company identify more at-risk customers and take preventive retention actions.

(c) Post-Pruning

A fully grown Decision Tree may overfit the training data.

Cost-complexity pruning controls tree complexity using:

[

$$R_\alpha(T)=R(T)+\alpha|T|$$

]

where:

- ($R(T)$) = error of the tree.
- ($|T|$) = number of terminal nodes.
- (α) = pruning parameter.

Example comparison:

Model	Accuracy
Pre-pruned Tree	90%
Post-pruned Tree	92%

Post-pruning removes unnecessary branches and generally improves generalization.

Recommended model: The **post-pruned Decision Tree** is more suitable for deployment if it provides better validation/test performance because it is less complex, easier to interpret, and less prone to overfitting.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is suitable because churn is a binary outcome.

It estimates:

[

$$P(Churn=1)=\frac{1}{1+e^{-z}}$$

]

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n$$

An illustrative comparison is:

Model	**Accuracy**	**AUC-ROC**
Decision Tree	92%	0.91
Logistic Regression	90%	0.94

AUC-ROC measures how well the model separates churners from non-churners.

When Logistic Regression is preferable

Logistic Regression is preferable when:

- Interpretability is important.
- The relationship between features and churn is reasonably structured.
- Probability estimates are required.
- The company needs to understand how each feature affects churn risk.
- A simpler and computationally efficient model is preferred.

(b) Feature Importance Analysis

Decision Tree

Suppose the three most important features are:

1. **Last Purchase Date**
2. **Purchase Frequency**
3. **Complaint History**

Logistic Regression

The three strongest predictors based on standardized coefficient magnitude may be:

1. **Last Purchase Date**
2. **Complaint History**
3. **Purchase Frequency**

Comparison

Both models identify the same three major factors, although their rankings may differ.

This agreement increases confidence that these features contain meaningful information about customer churn.

Interpretation:

- Long periods without purchasing indicate declining engagement.
- Frequent complaints indicate dissatisfaction.
- Low purchase frequency indicates weakening customer activity.

Task 4: Reinforcement Learning for Action Recommendation

(a) MDP Model

The retention recommendation problem can be modeled as a **Markov Decision Process (MDP)**.

States

Customer engagement can be represented using states such as:

- **S1 – Highly Engaged**
- **S2 – Moderately Engaged**

- **S3 – Low Engagement**
- **S4 – At Risk**
- **S5 – Churned/Inactive**

These states summarize the customer's current engagement level.

Actions

The agent can choose:

- **A1 – Discount**
- **A2 – Loyalty Offer**
- **A3 – Personal Outreach**
- **A4 – No Action**

Reward Function

A possible reward structure is:

Outcome	**Reward**
Customer retained	+10
Customer engagement improves	+5
Customer remains unchanged	+1
Customer churns	-10
Unnecessary costly offer	-2

The reward encourages the agent to retain customers while avoiding unnecessary incentives.

Transition Dynamics

After an action is taken, the customer's engagement state may change.

For example:

At Risk + Personal Outreach → Moderately Engaged
At Risk + Discount → Highly Engaged
Low Engagement + No Action → At Risk
At Risk + No Action → Churned

The transitions can initially be estimated from historical customer behavior and later improved using observed interactions.

(b) Q-Learning

Q-Learning learns the value of taking an action in a particular state.

The update equation is:

$$
Q(s,a)\leftarrow Q(s,a)+\alpha[r+\gamma\max Q(s',a')-Q(s,a)]
$$

where:

- α = learning rate
- γ = discount factor
- r = reward
- s' = next state

Assume:

$$
\alpha=0.5,\quad \gamma=0.9
$$

Initially all Q-values are zero.

Iteration 1

For:

State = At Risk, Action = Personal Outreach

Suppose reward = +5 and next state = Moderately Engaged.

[
Q(S4,A3)=0+0.5[5+0.9(0)-0]
]

[
Q(S4,A3)=2.5
]

For:

State = At Risk, Action = Discount

Suppose reward = +10.

[
Q(S4,A1)=0+0.5[10]=5
]

Iteration 2

Suppose the agent again selects **Discount** from At Risk and receives +10. Assume the maximum Q-value in the next state is 4.

[

$Q(S4,A1)=5+0.5[10+(0.9\times4)-5]$

]

Low Engagement + Loyalty Offer

Suppose:

- Current Q = 0
- Reward = +5
- Maximum next-state Q = 2

[

$Q(S3,A2)=0+0.5[5+(0.9\times2)]$

]

[

$Q(S3,A2)=3.4$

]

Example Q-Table

State	**Discount**	**Loyalty Offer**	**Personal Outreach**	**No Action**
Highly Engaged	1.0	2.0	1.0	3.0
Moderately Engaged	3.0	4.0	3.5	2.0
Low Engagement	2.0	**3.4**	4.5	1.0
At Risk	**6.8**	3.0	**2.5**	-4.0

The agent is trained for **at least 5 customer episodes**, with each episode consisting of customer state → action → reward → next state transitions.

Convergence Criterion

Training can be considered converged when:

$$
\max_{s,a}|Q_{new}(s,a)-Q_{old}(s,a)|<\epsilon
$$

for example, when the maximum Q-value change remains below **0.01** for several consecutive episodes.

(c) Final Learned Policy

The policy chooses the action with the highest Q-value:

$$
\pi(s)=\arg\max_a Q(s,a)
$$

Example:

Customer State	**Best Action**
Highly Engaged	No Action
Moderately Engaged	Loyalty Offer
Low Engagement	Personal Outreach
At Risk	Discount
Churned	Personal Outreach/Reactivation

Ethical Considerations

1. **Fairness** – Offers should not unfairly discriminate against customers based on sensitive or protected characteristics.

2. **Manipulation Risk** – The system should not aggressively pressure customers into purchases.
3. **Transparency** – Customers should understand why personalized offers or communications are being provided.
4. **Privacy** – Customer purchase and behavioral data must be securely handled.
5. **Cost and Incentive Fairness** – The system should avoid giving systematically worse offers to particular customer groups without valid business justification.
6. **Human Oversight** – High-impact retention decisions should remain reviewable by human staff.

Therefore, the RL agent should optimize **customer retention and long-term satisfaction**, rather than maximizing short-term revenue alone.